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Typology of laws restricting access to methadone treatment in the United States: A latent class analysis

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Abstract

Background: In the United States, methadone treatment for opioid use disorder is only available at opioid treatment programs (OTPs). In addition to federal regulations, states can enact laws which shape access to OTPs. We aimed to define classes of states according to restrictiveness of state OTP laws and examine population characteristics associated with class membership.

Methods: A set of laws was extracted from a database of statutes and regulations governing OTPs in 49 states and the District of Columbia as of June 2021. Latent class analysis of

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Declaration of Competing Interest

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Supplementary materials

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CRediT authorship contribution statement

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laws was used to estimate the probability of class membership for each state. Class-weighted multinomial logistic regression analysis assessed state-level correlates of class membership and adjusted Relative Risk Ratio (aRRR) and 95% confidence intervals (95% CI) were generated.

Results: States ($n = 50$) were assigned to three classes; Class 1) High restrictiveness on patient experience, low restrictiveness on access to service ($n = 13$); Class 2) Medium restrictiveness on patient experience, high restrictiveness on access to service ($n = 14$); Class 3) Low restrictiveness on patient experience, low restrictiveness on access to service ($n = 23$). States with a higher probability of membership in Classes with higher restrictiveness had higher rates of unemployment (Class 1 vs Class 3, aRRR:1.24; 95%CI:1.06–1.45), and Black residents (Class 2 vs Class 3, aRRR:1.10; 95%CI:1.04–1.15), and lower likelihood of Medicaid coverage of methadone (Class 1 vs Class 3, aRRR:0.25; 95%CI:0.07–0.88). States with a higher probability of membership in Classes with higher restrictiveness also had higher rates of potential indicators for opioid use disorder treatment need, including rates of opioid dispensing (Class 1 vs Class 3, aRRR:1.06; 95%CI:1.02–1.10, Class 2 vs Class 3, aRRR:1.07; 95%CI:1.03–1.11) and HIV diagnoses attributed to injection (Class 1 vs Class 3, aRRR:3.92; 95% CI:1.25–12.22).

Conclusions: States with indicators of greater potential need for opioid use disorder treatment have the most restrictions, raising concerns about unmet treatment need.

Keywords

Opioid treatment program; Methadone; Opioid agonist treatment; Law; USA

Introduction

Globally, there are around 40 million people dependent on opioids with the United States (US) having the highest estimated prevalence of opioid dependence (Degenhardt et al., 2019). In 2021 there were an estimated 5.6 million people with opioid use disorder (meeting the DSM-5 criteria for heroin use disorder or prescription pain reliever use disorder) in the US (Substance Abuse & Mental Health Services Administration, 2022a). There were estimated to be 9.2 million people who had used illicit opioids in the year prior to 2020, equating to 3.3% of the national population (Substance Abuse & Mental Health Services Administration, 2022a). The baseline risk associated with use of illicit opioids has greatly increased in recent years due to contamination of the supply with illicit fentanyl (Mattson et al., 2021), threatening the health of people who use opioids and putting them at risk of fatal overdose (Ivins et al., 2020). Between 2020 and 2021, overdose deaths involving opioids in the US rose by 15% (CDC/National Center for Health Statistics, 2022).

Opioid agonist treatment, referring to treatment for opioid use disorder with methadone or buprenorphine, is associated with reductions in all-cause mortality, fatal overdose (Santo et al., 2021), injecting risk behaviours, and the risk of HIV and HCV infection (Degenhardt et al., 2019). A treatment gap has been identified in the US with only 13% of people with opioid use disorder having received opioid agonist treatment in 2019. Of those, 43% receive opioid agonist treatment in opioid treatment programs (OTPs) (Krawczyk et al., 2022) (90% methadone (Substance Abuse & Mental Health Services Administration, 2020)). All others receive buprenorphine (56%) or naltrexone (1%) in non-specialty and

office-based care (Krawczyk et al., 2022). Improved treatment retention has been found for methadone compared to sublingual buprenorphine (Degenhardt et al., 2023). Methadone for the treatment of opioid use disorder is only legally available in the US at specialised OTPs which are primarily available in urban areas (Calcaterra et al., 2019), leading to disparities in access (Joudrey et al., 2020).

Although OTPs are permitted to provide methadone, buprenorphine, or naltrexone, in practice many only provide methadone and approximately 90% of attendees receive methadone (Substance Abuse & Mental Health Services Administration, 2021). Reduced choice in treatment options presents a barrier to treatment adherence (Frank et al., 2021). This is especially concerning given emerging evidence that methadone is effective in the context of increasing fentanyl prevalence in the drug supply (Stone et al., 2018). In light of the improved health outcomes for people with opioid use disorder who are receiving opioid agonist treatment (Santo et al., 2021) and given documented lack of access to the medication (Krawczyk et al., 2022), research is needed to identify combinations of the statutes and regulations (hereafter “laws”) that may impose legal barriers to OTP access.

In the US, OTPs are required by federal law to meet complex regulatory requirements (Federal Opioid Treatment Standards, 2002). State-level laws that enact further restrictions on OTPs beyond those stipulated by federal law can pose an additional legal barrier that can prevent programs from opening or staying in operation (Johnson et al., 2018), and can make it more difficult for potential patients to access OTPs and to stay in treatment. Prior literature has identified a number of state laws which reduce access to methadone in OTPs. State laws which prohibit the operation of medication units (here referring to non-mobile satellite units which are separate from the OTP but provide the same services) decreases the number of possible dosing points therefore reducing access to methadone (Wu et al., 2022). Laws requiring OTPs to have pharmacy licensure increases burden on staffing and costs which impacts OTP operation, especially in rural areas where staffing is limited (Lister et al., 2020). State zoning restrictions can limit access to OTPs by making them less convenient or inaccessible (Amiri et al., 2018). Requirements for clients to have government ID can produce an additional financial barrier to accessing healthcare (LeBrón et al., 2018). There is limited evidence to support the prohibition of take-home doses to reduce the incidence of opioid-related deaths or improve retention in care (Saulle et al., 2017). Permitting the use of administrative discharge for drug use/missed dose increases the possibility of clinical abandonment (Williams & Bonner, 2020). Requiring a set schedule for counselling has been found to have no effect on retention in care compared to voluntary counselling, and is associated with lower quality of life (Schwartz et al., 2017). Laws which restrict medication dose level can prevent people from accessing the dose they need and impact retention (Perreault et al., 2015). There is no evidence to support laws requiring observed collection of urine sample as a way to improve treatment retention or reduce opioid-related mortality (Chalabianloo et al., 2022; McEachern et al., 2019). As the laws in the US vary widely by state, they are not easy to summarize but are key to understanding the ways in which people are prevented from accessing methadone.

Recent federal regulations have been proposed by the Substance Abuse and Mental Health Services Administration to expand access to treatment from OTPs (Substance Abuse &

Mental Health Services Administration, 2022b). This is a promising development but will not address state-imposed laws which are stricter than federal regulations and are a major driver of access to treatment. Laws that govern OTPs are highly variable between states, with one review identifying 89 different laws across the country (Jackson et al., 2020). Jackson et al. found that a higher number of OTP laws in a state was associated with opioid-related deaths and opioid-related emergency visits, but this study did not examine the characteristics of states with more versus fewer restrictions on OTP operation. By characterizing the types of regulatory models adopted across states, we can identify the combinations of OTP laws which are enacted together. Further, by examining the features of states with laws that place additional barriers to OTP access and operation, we can identify the contexts with the greatest barriers to OTP access, and better describe inequities in methadone access across the country. This study aimed to define a typology of OTP laws across U.S. states in 2021, and to describe the characteristics of states according to their “class” of OTP law. We used a data-driven approach, latent class analysis, to classify states’ OTP laws into latent classes that reflect combinations of OTP laws that are likely to be adopted at the same time. Identifying latent classes allowed us to reduce complex state-level patterns of laws into a simpler typology to explore the relationship between OTP laws and state-level characteristics.

Methods

Opioid treatment program laws

The set of laws governing OTPs in the US was compiled by drug policy experts at Pew Charitable Trusts (Hereafter “Pew”), an independent non-profit, non-governmental organization which aims to improve public policy by conducting rigorous analysis. The methodology has been described in a comprehensive report of the laws (Caudell-Feagan et al., 2022). FM, SD, and VB developed an initial codebook after reviewing both statutes and administrative codes for all states (except Wyoming) and the District of Columbia, using the software LexisNexis®. Wyoming was excluded because the state does not have any OTPs or corresponding laws. The initial selection of laws was drawn from Jackson et al. (2020) and the Prescription Drug Abuse Policy System “Requirements for Licensure and Operations of Medications for Opioid Use Disorder Treatment” (Center for Public Health Law Research at the Temple University, 2020). In addition to these sources, the Pew research team reviewed the section of statutes providing rule-making authority to the agency that promulgated regulations in the administrative code. If either regulations or statutes referred to Board of Pharmacy oversight, the Pew research team also examined the relevant sections of administrative code produced by that body. Data represents state rules codified in statute and administrative code as of June 2021. It does not reflect any temporary changes such as executive orders or policy statements which states may have issued due to the COVID-19 pandemic.

The research team identified 24 laws which governed OTP operation. Once all 24 laws codified in statute and administrative code had been extracted, the findings were verified with state opioid treatment authorities (the official charged with overseeing OTPs). Between June and August 2021, the Pew research team sent the results of each jurisdiction’s

regulatory review to their state opioid treatment authority. The Pew research team received responses from 37 states. If the state opioid treatment authority disagreed with the research team's findings, the Pew research team reviewed and either corrected the findings or explained the decision not to. In cases where they did not update findings, those discrepancies were either due to differences in definitions or because updated regulatory language was not in effect until after 1 June 2021, the cut-off date for this analysis.

Creation of a state typology of opioid treatment program laws

OTP laws can be broadly classed in two groups. Laws that restrict access to service may prevent services from opening or prevent a person from enrolling as a patient. Laws that restrict the patient experience affect people once they are attending an OTP and may reduce retention in treatment. A subsample of the Pew database of legal provisions for OTPs was selected based on literature which identified them as restricting access to treatment (Described in Introduction). Where there were multiple variables for the same area of law (e.g., Take-homes prohibited in the first 30 days and Take-homes prohibited in the first 90 days), correlation between the variables was assessed and one was selected for inclusion. The selection was made based on the principle of parsimony (Zhang, 2016) and the research team's consensus on the policy with greatest potential for impact on care utilization. Laws that were present or absent in less than 10% of states were excluded to ensure there was sufficient heterogeneity. Hypothesizing that the laws would overlap or cluster together in certain groups of states, the analysis sought to identify these qualitatively different subgroups, or latent classes. Once the set of laws was selected, states were grouped together according to how these laws cluster. Latent class analysis allows us to assess the probability of a state belonging to a class based on categorical variables (e.g. the presence of a law in a state). Latent class analysis provides parameters that can be useful for identifying distinct classes or groups of states, thus allowing us to summarize complex combinations of policy provisions and identify those states with the most restrictive policies across a range of features. The approach is preferable to alternatives such as cluster analysis because it is model based and allows the use of model fit statistics to understand how well the model fits the data (Nylund-Gibson & Choi, 2018). The statistical approach to identify these latent classes is described in Statistical Analysis.

Correlates of typology

Due to the impact of COVID-19 on systems of data collection and data quality, data on correlates were extracted for 2019 where possible. Correlates were grouped into three domains; sociodemographic characteristics, policy characteristics and health measures. Correlates and their sources are reported in Supplementary Table 1.

To assess if restrictive OTP laws were found in states with more marginalised populations, we considered states' sociodemographic characteristics. Variables were selected based on prior literature on the factors associated with engagement in methadone treatment; sex (Marsh et al., 2021), age (Amiri et al., 2018; Mancino et al., 2010), race (Gryczynski et al., 2011; Krawczyk et al., 2017; Mancino et al., 2010), employment (Socías et al., 2020), poverty (Hall et al., 2021), insurance coverage (Hall et al., 2021) and rurality (Calcaterra et al., 2019).

To assess if restrictive OTP laws were found in states with restrictive or conservative policy environments, we considered four key variables. State adoption of Medicaid expansion (a government program providing health insurance for most low-income adults to 138% of the federal poverty level) was previously found to be associated with lower rates of opioid overdose deaths (Kravitz-Wirtz et al., 2020) and so the analysis considered Medicaid expansion and inclusion of methadone in Medicaid coverage in the state. The number of restrictions in state regulations was included, as states with heavier regulatory burdens across all domains were hypothesised to also have more laws restricting OTPs. This was measured as total number of restrictions (sum of the number of times the words <shall, must, may not, required, prohibited> appear in the regulatory text) imposed by any of the jurisdiction's regulations for 2020 (McLaughlin et al., 2020). States with laws permitting medical marijuana were included as it was hypothesised to be indicative of a more liberal approach to regulation of controlled substances.

Finally, to assess if restrictive laws were evident in states with greater health needs, we considered health measures related to opioid use: prevalence of opioid use disorder, proportion of people who have opioid use disorder who are receiving opioid agonist treatment (Nielsen et al., 2017), opioid dispensing excluding products to treat opioid use disorder (Degenhardt et al., 2007), overdose (Krawczyk et al., 2017), and HIV infections attributable to drug use (Sullivan et al., 2005).

Statistical analysis

The primary aim of the study was to characterize states according to restrictive OTP laws at a discrete time point. We conducted exploratory latent class analyses using MPlus Version 8.8 (Muthén & Muthén, 1998) to select the optimal number of classes. A latent class model was fit with two through six classes. The model of best fit was chosen based on Akaike Information Criterion (AIC) (Akaike, 1974), Bayesian Information Criterion (BIC) (Schwarz, 1978), sample size adjusted BIC (SSABIC), and the extent to which the assigned classes were interpretable. Once the model of best fit was chosen, the item endorsement probabilities for each class were described, the characteristics of each class were described, and a map was used to present the states' most probable class assignment.

The secondary aim of the study was to evaluate the association between state characteristics and latent class membership for OTP laws. A class-weighted multinomial logistic regression analysis was performed using the Bolck Croon Hagenaars method (Asparouhov & Muthén, 2014). Possible correlates were identified *a priori* (See Methods: Correlates of Typology) and clustered in three groups: sociodemographic characteristics, policy environment, and health measures. Multinomial logistic regression analyses were used to determine the association between covariates and class membership, expressed as relative risk ratios with 95% confidence intervals. Covariates that reached statistical significance ($p < 0.05$) in the unadjusted models were included in the adjusted multinomial model. For the adjusted multinomial logistic regression model, the association between each cluster of covariates and latent class membership was assessed. Collinearity within a cluster (indicated by a variance inflation factor (VIF) of more than four) was investigated and the variable deemed to be more pertinent according to prior literature was retained in the adjusted model.

Results

Characteristics of state opioid treatment program laws

The complete set of laws compiled by the Pew research team has been reported elsewhere (Caudell-Feagan et al., 2022). This analysis included 11 types of laws which are implemented by more than 10% of states and restrict access to treatment or worsen the patient experience. The laws that were discarded because they are absent in less than 10% of states are presented in Supplementary Table 2. The final selection of 11 laws is presented in Table 1.

Latent class analysis

The model fit statistics indicated a preference for the three-class solution according to AIC, a preference for a two-class solution according to BIC and a preference for a five-class solution according to SSABIC (Supplementary Table 3). After assessing the class composition, the three-class solution was chosen based on ease of interpretability.

States were assigned based on their probability of class membership. For Class 1 (High restrictiveness on patient experience, low restrictiveness on access to service) 13 states were assigned including California and Florida with item endorsement probabilities highest for the requirement to perform additional drug tests (0.84), and requirement to follow a set schedule for counselling (0.84). For Class 2 (Medium restrictiveness on patient experience, high restrictiveness on access to service) 14 states were assigned including Georgia and New Jersey with item endorsement probabilities highest for the requirement for OTPs to have pharmacy licensure (0.79), and requirement to employ pharmacists on site (1.00). For Class 3 (Low restrictiveness on patient experience, low restrictiveness on access to service) 23 states were assigned including Texas and Illinois with item endorsement probabilities less than 0.30 across all laws (Fig. 1).

Fig. 2 shows the assigned Class membership alongside the state-level proportion of people with opioid use disorder who are receiving medication. States in the more restrictive Class 1 and Class 2 were concentrated in the South and Northeast Regions and states in the Northeast Regions had higher proportions of people receiving treatment. States in the less restrictive Class 3 were concentrated in the West and Midwest regions. Characteristics of the three classes are reported in Table 2.

Class-weighted multinomial regression analysis

Each cluster of variables in the multinomial regression model was analysed for Class 1 (High restrictiveness on patient experience, low restrictiveness on access to service) and Class 2 (Medium restrictiveness on patient experience, high restrictiveness on access to service) with reference group of Class 3 (Low restrictiveness on patient experience, low restrictiveness on access to service). Results are presented in Table 3.

Sociodemographic characteristics cluster

There was collinearity in this cluster, with high VIF for sex (VIF = 4.59), unemployment (VIF = 6.56), and poverty (VIF = 6.37), so it was not possible to include them together

in the same model. Unemployment was retained in the multivariate analysis because of literature supporting its importance in engagement in treatment (Azagba et al., 2021; Proctor et al., 2015). After removing the variables of sex and poverty, VIF was <2 for all remaining variables in the cluster. The only factor independently associated with membership in Class 1 vs Class 3 in the sociodemographic characteristics cluster was higher unemployment rates (aRRR 1.24, 95% CI: 1.06–1.45). Factors significantly associated with membership in Class 2 vs Class 3 in the sociodemographic characteristics cluster were higher median age (aRRR 1.31, 95% CI: 1.00–1.70), higher proportion of Black population (aRRR 1.10, 95% CI: 1.04–1.15), higher proportion of population that was uninsured (aRRR 1.27, 95% CI: 1.06–1.53), and lower proportion of urbanicity (aRRR 0.95, 95% CI: 0.92–0.99) (Table 3).

Policy environment cluster

Factors significantly associated with membership in Class 1 vs Class 3 in the policy environment cluster were likelihood of the state including methadone in Medicaid (aRRR 0.25, 95% CI: 0.07–0.88) and increased regulatory restrictions in all state regulations (aRRR 1.11, 95% CI: 1.05–1.18). The only factor significantly associated with membership in Class 2 vs Class 3 in the policy environment cluster was lower enactment of medical marijuana laws (aRRR 0.37, 95% CI: 0.15–0.94) (Table 3).

Health measures cluster

Factors significantly associated with membership of Class 1 vs Class 3 in the health measures cluster were higher rates of opioid dispensing (aRRR 1.06, 95% CI: 1.02–1.10) and higher rates of HIV diagnosis attributable to injection (aRRR 3.92, 95% CI: 1.25–12.22). The only factor significantly associated with membership of Class 2 vs Class 3 in the health measures cluster was higher rates of opioid dispensing (aRRR 1.07, 95% CI: 1.03–1.11) (Table 3).

Discussion

This study applied latent class analysis to characterize the complex landscape of OTP laws across individual states in the US. This study finds states with greater need for access to methadone have more restrictive OTP laws. States' regulatory approaches to OTPs can be classified into three types: states with minimal restrictions in addition to federal laws, states with high restrictiveness on the patient experience, and states with medium restrictiveness on the patient experience and high restrictiveness on access to services. Under US federal law, methadone for opioid use disorder is only available from OTPs (Joudrey et al., 2021) so by investigating barriers to OTP access this study effectively investigates barriers to methadone access. The inconvenience of attending an OTP is likely exacerbated by the additional state-level laws described in this study, which prevent low threshold treatment provision. This analysis demonstrates the ways in which groups of states add to the extensive federal restrictions, increasing requirements for people and OTPs that may generate barriers to accessing methadone and widen inequities in health.

The sociodemographic correlates of membership in classes with higher restrictions on OTPs reveal potentially concerning sources of inequity in treatment access. For example, states

with higher unemployment rates were more likely to be classified as high restriction on patient experience. People who are unemployed already face barriers to care including the out-of-pocket costs associated with OTP attendance (Dunphy et al., 2021). States with a larger Black population, a higher proportion of uninsured people, and lower urbanicity were more likely to belong to the class with high restriction on access to services. In a population-level study in the US, Medicaid enrollees receiving methadone and buprenorphine increased at a lower rate in counties with a higher concentration of Black and Hispanic people (Stein et al., 2018). Having more restrictions on OTPs in states with a higher proportion of Black people risks entrenching the structural racism in treatment access (Peterkin et al., 2021). Further, concentration of more restrictive OTP laws in states with lower urbanicity could contribute to the already low availability of opioid agonist treatment providers in rural areas (Haffajee et al., 2019). Reducing restrictions on OTPs may facilitate the opening of new services in underserved areas.

In the policy environment cluster, membership in the class with high restrictiveness on access to services was associated with states not including methadone in Medicaid coverage. If methadone is not covered by Medicaid, out of pocket fees can present a barrier to accessing methadone for low-income populations. More broadly, expansion of Medicaid has been found to correlate with reduction in total opioid overdose deaths (Kravitz-Wirtz et al., 2020), demonstrating its importance for people who use opioids. States need to consider how the regulatory environment can support OTPs to provide low threshold access to methadone, particularly in response to the increased prevalence of fentanyl in the drug supply. Enactment of medical marijuana laws was negatively associated with membership of the class with high restrictiveness on access to services, suggesting more permissive drug regulation might also be indicative of more permissive policy environments regarding OTPs.

In the health measures cluster, the association between the state's rate of HIV infections attributable to injecting drugs and membership in the class with high restrictiveness on patient experience indicates that more restrictive laws are enacted in states with greater public health need for methadone access. Methadone has been associated with a 54% reduction in HIV transmission (MacArthur et al., 2012), supporting the need to improve methadone access in areas where there are more HIV diagnoses. Additionally, opioid dispensing was associated with membership in Classes with more restrictive laws. Evidence on using methadone to treat pharmaceutical opioid dependence is limited but does indicate improved retention compared to buprenorphine (Nielsen et al., 2017). Reducing restrictions on OTPs can lower barriers to care to ensure everyone that needs methadone can access it.

Some of the most common state laws require pharmacy licensure/pharmacist services, additional drug tests, or a set schedule for counselling. Federal law already requires OTPs to have a licence to operate, so requiring an additional pharmacy licence adds a further bureaucratic requirement which could prevent OTPs from opening or operating. Similarly, federal law already requires that only licenced healthcare practitioners provide methadone so the requirement that OTPs employ a pharmacist is an additional bureaucratic step and burden on resources which could prevent OTPs from opening or operating. Given that the lack of staff is already a barrier to access treatment in rural areas (Lister et al., 2020), additional requirements for staffing would likely worsen issues with access. Regarding

additional drug tests and set schedules for counselling, observed urine drug screenings risk impacting the development of therapeutic relationships (Strike & Rufo, 2010) and models of care which involve compulsory counselling may not result in improved patient outcomes when compared to methadone treatment with voluntary counselling (Schwartz et al., 2017) or compared to models of shared decision-making (Marshall et al., 2022). These examples of low-value care should be removed from the law in order to improve the patient experience.

This study highlights the need to look beyond federal or national laws, to gain a comprehensive understanding of the legislative environment which drives treatment uptake. Methadone remains stigmatised and difficult to access, despite evidence to support its use as a first line response for people who are seeking treatment (Degenhardt et al., 2023; Mattick et al., 2009). Globally, there is a gap between those who may need opioid agonist treatment and those who are accessing it, particularly outside of Australasia and Western Europe (Colledge-Frisby et al., 2023). The analytical approach outlined in this study can be used in other settings to explore the complex legislative environments which govern treatment access, to understand which laws are enacted in tandem and better identify areas with significant legal barriers to treatment access despite high treatment need. For countries where methadone is available, future research can build on this study to investigate how restrictive legislative environments may be shaping access to methadone and treatment-related outcomes, including opioid use disorder and overdose.

Consistent and evidence-based laws could increase equitable access to methadone in the US and reduce the gap between people who would benefit from opioid use disorder treatment and those who actually receive it (Krawczyk et al., 2022). In the US, the legal restrictions outlined in this study serve to mandate certain treatment models and limit the provision of individualised, person-centered care for patients with opioid use disorder. Restrictions were put in place in part out of a concern that broader, less regulated access to methadone could result in increased unprescribed use and involvement of methadone in overdose. However, the beginning of the COVID-19 pandemic saw increased flexibilities for take-home methadone doses and, as yet, no significant increases in methadone overdoses have been observed (Krawczyk et al., 2023). Evidence from this period is useful to bolster previous studies which found that less restrictive approaches to methadone provision are as effective at reducing drug-related harms (Nolan et al., 2015), improving initiation (Carter et al., 2022) and improving retention in treatment (Perreault et al., 2015). Recent proposed changes to federal regulations will give more autonomy to OTP clinicians (Substance Abuse & Mental Health Services Administration, 2022b), but this will not address the state-level laws which remain an important driver of methadone access. Choice and flexibility in treatment options must be prioritised to ensure people can access methadone when they need it. Studies suggest that treatment retention is better for methadone than sublingual buprenorphine (Degenhardt et al., 2023) and so low threshold access to both methadone and buprenorphine is required to support the response to the opioid overdose crisis. This could be facilitated by removing restrictions on OTPs and moving methadone dispensation to office-based and pharmacy dispensing models (Adams et al., 2022).

There are several limitations to this study. Laws that were present or absent in less than 10% of states were discarded, despite restricting access to methadone, to ensure there was

sufficient heterogeneity to inform the latent class analysis. For example, West Virginia had the highest rate of opioid overdose in 2020 (Centers for Disease Control and Prevention, National Center for Health Statistics 2021), yet was the only state with a moratorium on opening new OTPs (Caudell-Feagan et al., 2022). Pennsylvania prohibits the operations of medication units (mobile or non-mobile satellite units which are separate from the OTP but can provide the same services) (Caudell-Feagan et al., 2022), despite a scoping review finding that medication units improve retention in care and improve access to methadone for underserved populations (Chan et al., 2021). Although not included in the proposed typology, these restrictive laws may limit access to lifesaving treatment where they are needed most. Performing a latent class analysis at the state level with a sample size of only 50 weakens the power of the analysis (Dziak et al., 2014), but the same data is not available at the county or city level. Nevertheless, the identification of three distinct Classes (with high entropy of 0.935 (Celeux & Soromenho, 1996)) and a categorisation which aligns with empirical data on the laws in each state can provide confidence that the typology is useful to facilitate further analysis like the multinomial regression presented. There was no investigation of the directionality of the associations between correlates and class membership, as it was a preliminary and exploratory analysis with no suggestion of causality. There is likely high variation in how the laws are implemented in practice and the study fails to acknowledge the “interpretive acts of individual health care workers” (Fraser & valentine, 2008), clinics, and local rules which impact enactment of laws. All laws that were included in the analysis were treated as equal when they likely differ in terms of impact on patient experience and access to treatment.

There is an abundance of federal and state laws governing operation of OTPs in the US. This study focuses on 11 that are commonly applied at the state-level and may contribute to restricting OTP access. The proposed typology advances understanding of the regulatory models for treatment provision in the US. An exploratory examination of the correlates of latent class membership reveals a concentration of more restrictive OTP regulatory approaches in states with populations that have historically had poorer access to healthcare, raising concerns about OTP restrictions running counter to public health goals and further exacerbating socioeconomic and racial inequalities in treatment access. Removing restrictive laws could facilitate access to OTPs and therefore methadone, which is of timely importance during a national overdose epidemic.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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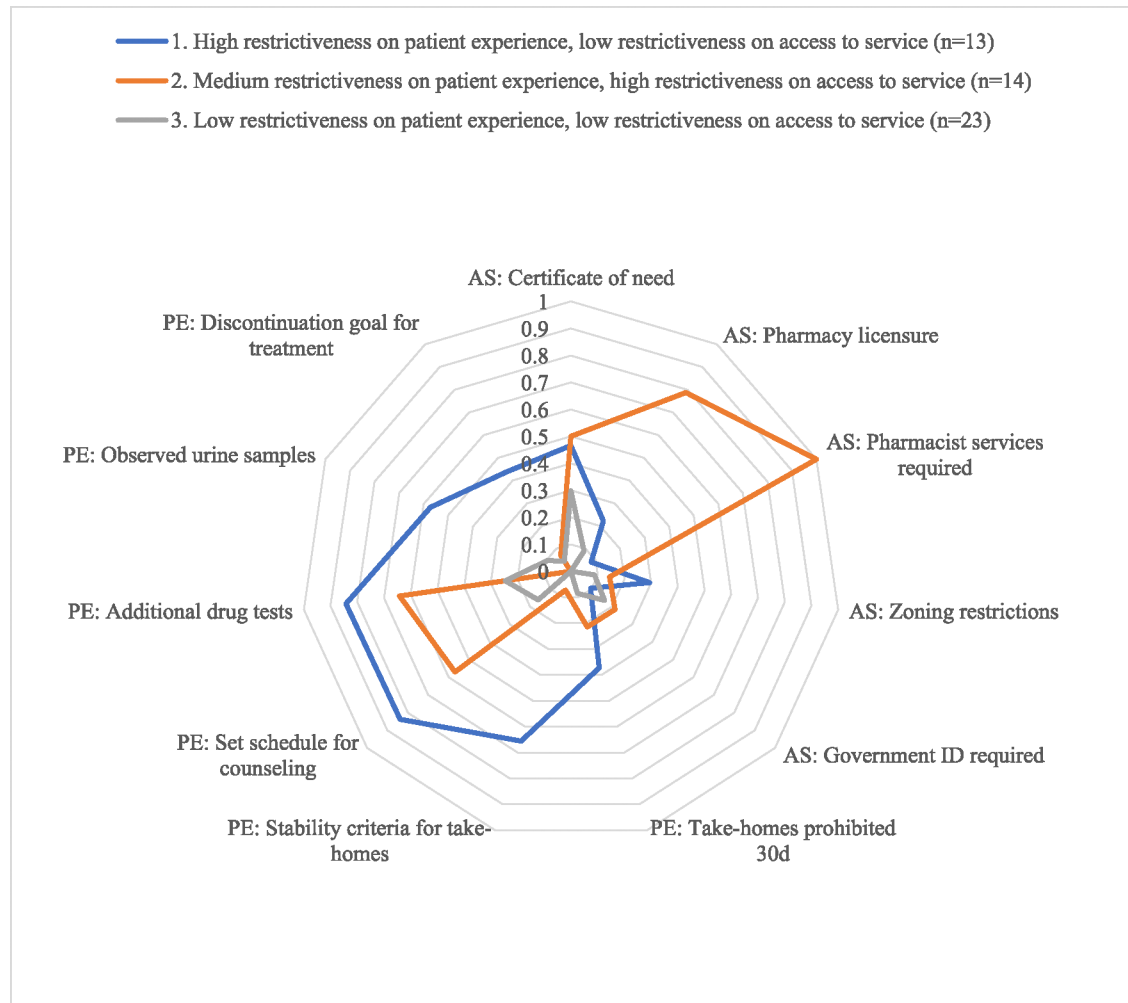
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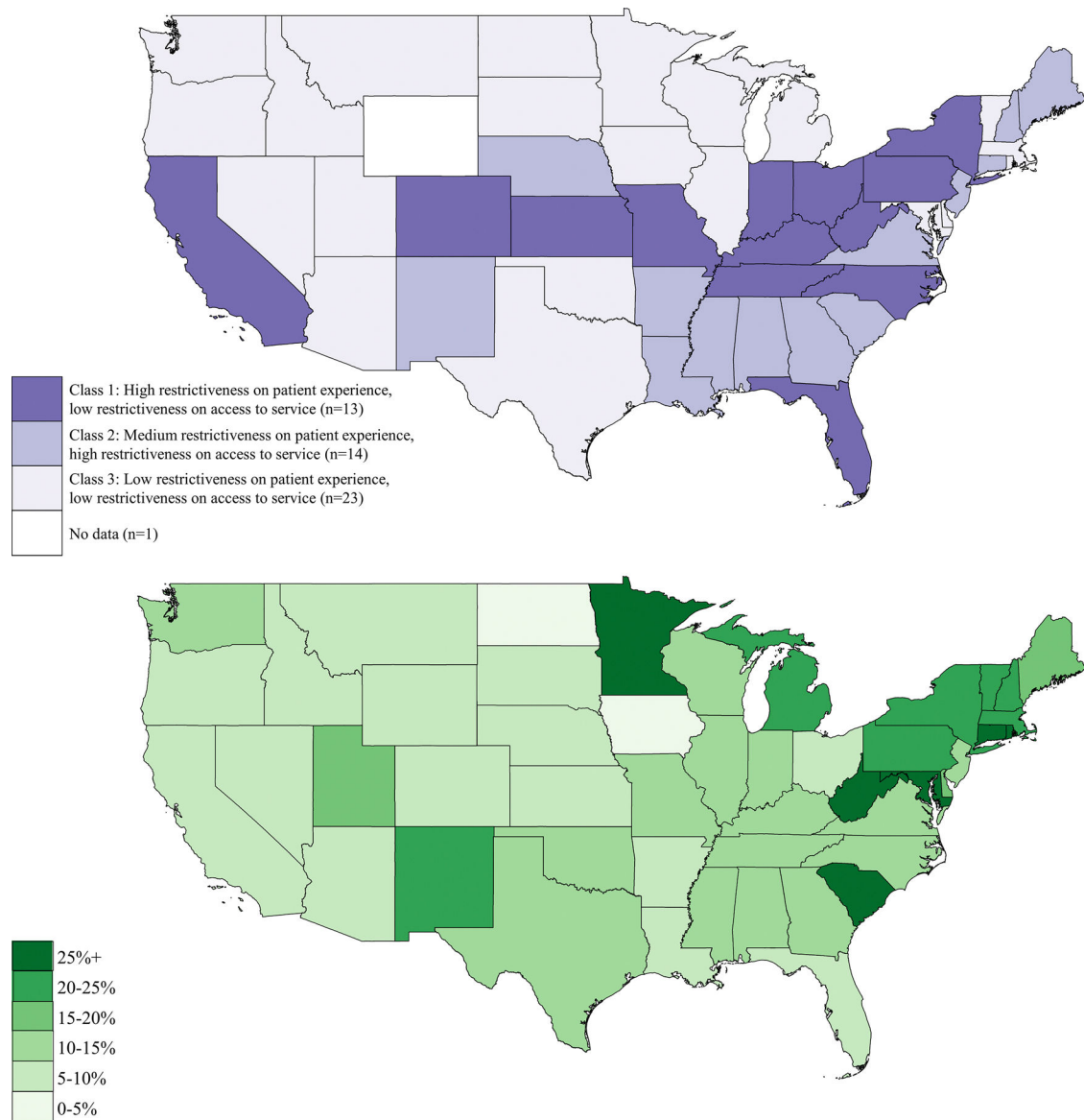
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**Fig. 1.**

Item endorsement probabilities by latent class among 50 states, US, 2021.

See Table 1 for full description of regulations.

Acronyms: AS: Access to services; PE: Patient experience.

**Fig. 2.**

a: Map of latent class membership by state in US – opioid treatment program regulations that restrict access to treatment, 2021

Composition of latent class membership as follows: Class 1: California, Colorado, Florida, Indiana, Kansas, Kentucky, Missouri, New York, North Carolina, Ohio, Pennsylvania, Tennessee, West Virginia. Class 2: Alabama, Alaska, Arkansas, Connecticut, Georgia, Louisiana, Maine, Mississippi, Nebraska, New Hampshire, New Jersey, New Mexico, South Carolina, Virginia. Class 3: Arizona, Delaware, District of Columbia, Hawaii, Idaho, Illinois, Iowa, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, North Dakota, Oklahoma, Oregon, Rhode Island, South Dakota, Texas, Utah, Vermont, Washington, Wisconsin

b: Map of proportion of people with opioid use disorder receiving medications for opioid use disorder by state in US – 2019.

Proportion calculated: Rate of individuals receiving medications for opioid use disorder (Individuals dispensed medications in opioid treatment programs and National Survey of Substance Abuse Treatment Services and individuals dispensed buprenorphine from IQVIA) divided by the rate of adjusted opioid use disorder prevalence (From the National Survey on Drug Use and Health, adjusted using Keyes et al. 2022 multiplier method).

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Table 1

Opioid treatment program (OTP) regulations included in the latent class analysis (US, June 2021).

Regulation - description	States (n/N)	Negatively impacts
Certificate of need required - States are interpreted to require a Certificate of Need if programs are required to demonstrate need prior to approval, even if the exact term “certificate of need” is not used.	20/50	Access to services
Pharmacy licensure or registration required - State requires a license from or registration with the Board of Pharmacy to operate an OTP	16/50	Access to services
Pharmacist services required - State requires the employment of a pharmacist at OTPs or the use of a consultant pharmacist	15/50	Access to services
State zoning restrictions - State places restrictions on where an OTP can operate	8/50	Access to services
Government ID required to access treatment - State requires a patient to have a government ID to be admitted	8/50	Access to services
Take-homes prohibited in the first 30 days - State prohibits take-home medications in the first 30 days of treatment	10/50	Patient experience
Additional stability criteria for take-homes - State has a specific definition of “stability” for takehomes beyond what is described in federal regulations	10/50	Patient experience
Set schedule for counseling sessions - State specifies a set counseling schedule with a minimum number of sessions based on time in treatment or other factors (Includes requiring a set schedule to receive take-homes)	23/50	Patient experience
Additional drug tests required - State requires drug testing more frequently than federal rules (at least 8 per year)	26/50	Patient experience
Observed collection of urine samples required - State has a blanket requirement that OTPs observe urine specimen collections	10/50	Patient experience
Discontinuation of medications as a goal for treatment - State establishes the discontinuation of medication as the goal of treatment (Includes states that require clients to be offered withdrawal management at regular intervals)	8/50	Patient experience

Characteristics of latent classes of opioid treatment program regulations that restrict access to treatment in 50 states in the US, 2021.

Table 2

	Class 1: High restrictiveness on patient experience, low restrictiveness on access to service (n = 13)	Class 2: Medium restrictiveness on patient experience, high restrictiveness on access to service (n = 14)	Class 3: Low restrictiveness on patient experience, low restrictiveness on access to service (n = 23)
Sociodemographic characteristics			
Female, median% (IQR)	51 (51–51)	51 (51–51)	50 (50–51)
Median age, median years (IQR)	39 (38–40)	38 (37–40)	38 (37–40)
Black, median% (IQR)	11 (6–15)	14 (3–27)	6 (2–12)
Unemployed, median% (IQR)	37 (36–39)	37 (34–42)	35 (32–37)
Living in poverty, median% (IQR)	13 (12–13)	12 (10–16)	11 (10–13)
Uninsured, median% (IQR)	8 (6–9)	9 (8–10)	6 (5–10)
Urbanicity, median% of counties in state (IQR)	74 (66–86)	70 (59–76)	83 (66–91)
Medicaid expansion, n(%)	8 (62)	8 (57)	17 (74)
Medicaid coverage includes methadone, n(%)	10 (77)	10 (71)	21 (91)
State regulatory restrictions, median 1000s (IQR)	12 (11–17)	12 (10–14)	10 (7–16)
Medical cannabis laws enacted, n(%)	8 (62)	8 (57)	18 (78)
Opioid use disorder, per 1000 pop (IQR)	8 (6–9)	6 (4–9)	8 (5–9)
People receiving buprenorphine at pharmacy for opioid use disorder, median per 1000 pop (IQR)	3 (1–4)	3 (2–3)	2 (1–3)
People receiving OAT in OTP, median per 1000 pop (IQR)	1 (1–2)	2 (1–3)	1 (1–2)
Opioid dispensing, median per 100 pop (IQR)	57 (45–60)	47 (41–67)	43 (40–50)
Opioid overdose, median per 100,000 pop (IQR)	9 (7–10)	8 (5–10)	8 (5–10)
HIV diagnosis attributable to injection, median per 100,000 pop (IQR)	1 (1–2)	1 (0–1)	1 (0–1)
Health outcomes			

* Restrictions defined as total number of restrictions (sum of the number of times the words <shall, must, may not, required, prohibited> appear in the regulatory text) imposed by any of the jurisdiction's regulations for 2020 (McLaughlin et al., 2020).

Table 3

State level correlates of membership in latent classes representing more restricted access to opioid treatment program regulation in 50 states in the US, 2021 (relative to membership in a latent Class 3 denoting low restrictiveness on patient experience and access to services).

	Class 1: High restrictiveness on patient experience, low restrictiveness on access to service		Class 2: Medium restrictiveness on patient experience, high restrictiveness on access to service	
	cRRR (95% CI)	aRRR (95% CI)	cRRR (95% CI)	aRRR (95% CI)
Sociodemographic				
Female,%	1.73 (1.01–2.95)	-	1.80 (1.06–3.06)	-
Median age, years	1.21 (1.02–1.45)	1.14 (0.88–1.49)	1.19 (1.00–1.41)	1.31 (1.00–1.70)
Black,%	1.02 (0.98–1.07)	1.02 (0.97–1.07)	1.07 (1.03–1.11)	1.10 (1.04–1.15)
Unemployed,%	1.29 (1.14–1.47)	1.24 (1.06–1.45)	1.25 (1.11–1.42)	1.01 (0.86–1.19)
Living in poverty,%	1.29 (1.09–1.53)	-	1.24 (1.05–1.46)	-
Uninsured,%	1.09 (0.94–1.25)	1.06 (0.88–1.27)	1.22 (1.06–1.40)	1.27 (1.06–1.53)
Urbanicity,% of counties	0.99 (0.96–1.01)	1.00 (0.97–1.04)	0.96 (0.93–0.98)	0.95 (0.92–0.99)
Policy environment				
Medicaid expansion, vs none	0.56 (0.24–1.31)	-	0.47 (0.21–1.06)	-
Medicaid coverage includes methadone	0.32 (0.10–0.97)	0.25 (0.07–0.88)	0.24 (0.08–0.70)	0.43 (0.12–1.48)
State regulatory restrictions *, 1000s	1.08 (1.02–1.14)	1.11 (1.05–1.18)	1.01 (0.95–1.08)	1.04 (0.97–1.11)
Health measures				
Medical marijuana laws enacted	0.44 (0.19–1.05)	0.44 (0.16–1.18)	0.37 (0.16–0.86)	0.37 (0.15–0.94)
Opium, per 1000 pop	1.06 (0.95–1.19)	-	0.93 (0.82–1.05)	-
People receiving buprenorphine at pharmacy for OUD, per 1000 pop	1.33 (1.05–1.67)	1.14 (0.87–1.51)	1.25 (0.99–1.57)	1.24 (0.98–1.58)
People receiving OAT in OTPs, per 1000 pop	0.73 (0.53–1.01)	-	0.91 (0.72–1.16)	-
Opioid dispensing, per 100 pop	1.06 (1.03–1.10)	1.06 (1.02–1.10)	1.07 (1.03–1.11)	1.07 (1.03–1.11)
Opioid overdose, per 100,000 pop	1.08 (0.97–1.20)	-	1.04 (0.93–1.16)	-
HIV diagnosis attributable to injection, per 100,000 pop	4.89 (1.65–14.48)	3.92 (1.25–12.22)	1.55 (0.54–4.47)	1.03 (0.33–3.23)

Acronyms - cRRR: crude relative risk ratio; aRRR: adjusted relative risk ratio; CI: Confidence intervals; OUD: opioid use disorder; OTP: opioid treatment program.

Bold indicates statistically significant RRR with p value<0.05.

* Restrictions defined as total number of restrictions (sum of the number of times the words <shall, must, may not, required, prohibited> appear in the regulatory text) imposed by any of the jurisdiction's regulations for 2020 (McLaughlin et al., 2020).

Source of predictors listed in Supplementary Table 1.